

Athul Suresh

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EDUCATION

Texas A&M University

Aug. 2022 – May 2026 (Expected)

Bachelor of Science in Applied Mathematics, Minor in Computer Science

College Station, TX

- Relevant Courses: Data Structures and Algorithms, Linear Algebra, Programming Languages, Discrete Math, Design & Analysis of Algorithms, Multivariate Calculus, Principles of Statistics I & II

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, Typescript, SQL (Postgres), R

Frameworks/Libraries: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib, Hugging Face, LangChain

Developer Tools: Git, Docker, Unix, Github, Jupyter Notebooks, Amazon Elastic Beanstalk, Jira, Postman, VS Code

Expertise: Machine Learning, Deep Learning, NLP, Reinforcement Learning, RAG Systems, Prompt Engineering, Statistical Modeling, Data Preprocessing

EXPERIENCE

ML Technical Lead

Jan. 2024 – Present

Software Subteam, TAMU Formula Electric

College Station, TX

- Optimized LSTM and Random Forest models for voltage sag prediction using PyTorch and Scikit-learn, achieving **40% accuracy improvement** through hyperparameter tuning and feature engineering on time-series data.
- Deployed real-time NLP system using Groq API and Llama 3.1 for automated order processing, achieving **sub-100ms inference** on production SQL databases with 95% extraction accuracy.

Software Engineering Intern

May 2025 – Aug. 2025

Ziff Davis Shopping

Austin, TX

- Engineered AI customer service chatbot with custom RAG workflows and prompt engineering, reducing query response time by 60% and automating support for **millions of enterprise users**.
- Developed GraphQL APIs processing 10k+ daily authentication requests with TypeScript and Node.js, implementing data pipelines for user management across **live production systems**.

Research Volunteer

Nov. 2024 – May. 2025

PiStar Lab, Texas A&M University

College Station, TX

- Implemented least-squares temporal-difference learning algorithm in RESCO traffic simulation using TensorFlow, achieving **15% reduction in average wait times** across multi-intersection scenarios.
- Validated RL-based traffic control on real-world datasets with **10k+ vehicle trajectories**, ensuring algorithm robustness and RESCO toolkit integration.

Technical Co-founder

April 2023 – Aug. 2023

Preptify

Austin, TX

- Implemented **language model fine-tuning** techniques such as few-shot learning which improved response relevance, and developed ETL pipelines for efficient data processing and analysis.
- Developed a Firestore-based analytics system and used A/B testing to **optimize platform performance**.

Software Engineering Intern

May 2021 – Aug. 2021

RetailMeNot

Austin, TX

- Engineered high-performance RESTful API using TypeScript and Node.js, **reducing data retrieval time by 70%** and accelerating project timeline by 3 weeks through standardized endpoint structures.

PROJECTS

Hermes AI Call Center | *Next.js, OpenAI, Twilio, PostgreSQL, Prisma*

Jan. 2025

- Architected scalable AI call center using OpenAI's Realtime Voice API, enabling businesses to **automate thousands of customer interactions** through intelligent conversation flows and context-aware responses.
- Engineered **concurrent call processing system** managing multiple simultaneous conversations with automated human escalation and real-time analytics dashboard providing live transcriptions and sentiment analysis.

AlphaZero TicTacToe Implementation | *Tensorflow, Numpy, Matplotlib, Git*

March 2024 – April 2024

- Implemented AlphaZero algorithm from scratch using TensorFlow with **50,000+ self-play games**, incorporating Monte Carlo Tree Search, neural network policy/value functions, achieving superhuman performance through RL.
- Achieved 0% loss rate, demonstrating ability to **recreate research papers** and apply cutting-edge algorithms.